

2. DETERMINATION OF WAVELENGTH OF SPECTRAL LINE OF MERCURY SPECTRUM USING DIFFRACTION GRATING - NORMAL INCIDENCE

Aim: To determine the wavelength of a different spectral lines of a given source of light by using the diffraction grating in the normal incidence position.

Apparatus: Plane diffraction grating, spectrometer, spirit level, reading lens, sodium vapour lamp.

Description: A plane diffraction grating consists of a parallel-sided glass plate with equidistant fine parallel lines drawn very closely upon it by means of a diamond point. The number of lines drawn is about 15,000 per inch.

Theory: When light of wavelength λ is incident normally on a diffraction grating having N lines per cm and if θ is the angle of diffraction in the n^{th} order spectrum, then

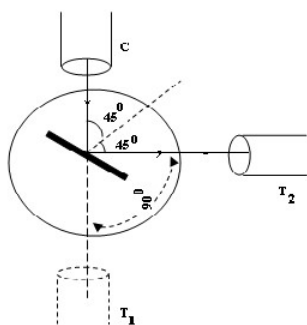
$$n N \lambda = \sin \theta$$

$$\text{or, } \lambda = \frac{\sin \theta}{N n}$$

from which λ can be determined.

Procedure: The usual initial adjustments of the spectrometer (Re Exp 18) are done. The least count of the vernier of the spectrometer is found.

- 1) Normal incidence:** The slit of the spectrometer is illuminated with sodium vapour lamp. The telescope is placed in line with the axis of the collimator and the direct image of the slit is observed. The slit is narrowed and the vertical cross-wire is made to coincide with the centre of the image of the slit (T_1 in fig.) The reading of one of the verniers is noted. The prism table is clamped firmly and the telescope turned through exactly 90° and fixed in position (T_2 in fig.)

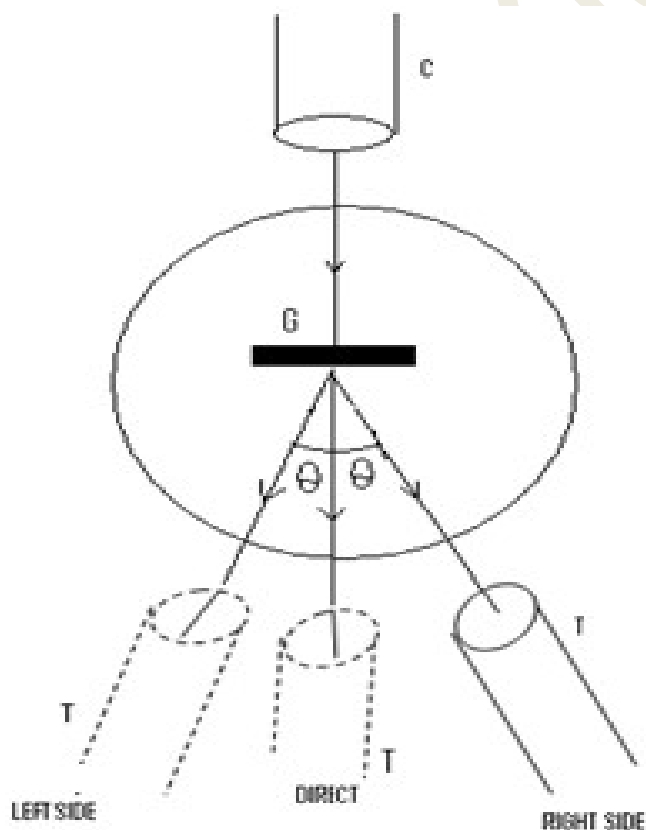


The grating is held with the rulings vertical and mounted in its holder on the prism table such that the plane of the grating passes through the centre of the table and the ruled surface towards the collimator. The prism table is released and rotated until the image of the slit is seen in the telescope by reflection on the ruled side of the grating. The prism table is fixed after adjusting the point of intersection of the cross-wires is on the image of the slit. Then the vernier table is released and rotated through exactly 45° from this position so that the ruled side of grating faces the collimator. The vernier table is fixed in this position and the telescope is brought back to the direct reading position. Now the light from the collimator strikes the grating normally.

2) **Measurement of λ :** The telescope is so as to catch the first order directed image on one side, say on the left (fig.) With sodium light two images of slit, very close to each other, can be seen. They are called D_1 and D_2 lines. The point of intersection of the cross wires is set on the D_1 line and its reading is noted on both the verniers similarly the reading corresponding to the D_2 line is noted. Then telescope is turned to the other side (i.e.), right side and similarly the reading corresponding to D_1 and D_2 lines of the first order spectrum are noted. Half the difference in the reading corresponding to any one line gives the angle of diffraction (θ) for that lines in the first order spectrum.

The experiment is repeated for the second order spectrum. The number of lines per cm. of the grating (N) is noted and the wavelength λ of the spectral line is found by the relation.

$$\lambda = \frac{\sin \theta}{Nn}$$



$$\text{Least count of the Spectrometer} = \frac{\text{value of 1MSD}}{\text{Total no. of divisions on the vernier scale}} = \frac{\frac{1^\circ}{2}}{30} = \frac{30'}{30} = 1'$$

Tabular form:

Direct readings:

 $V_1' =$ $V_2' =$

Colour	Reading of spectrometer		Difference		Average(θ) $\frac{\theta_1 + \theta_2}{2}$	Sin θ	Wavelength (λ) Å	
	Vernier (V_1) MSR+(V.C X L.C)	Vernier (V_2) MSR+(V.C X L.C)	$\theta_1 = V_1 \sim V_1'$	$\theta_2 = V_2 \sim V_2'$			Calculated	Standard
Violet - I								4047
Violet - II								4078
Blue								4358
Bluish-green								4916
Green								5461
Yellow-I								5890
Yellow-II								5896
Red								6234

Calculations:

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Precautions:-

- 1) Always the grating should be held by the edges.
- 2) The rule surface should not be touched.
- 3) Light from the collimator should be uniformly incident on the entire surface of the grating.

Result:- The wavelength of various spectral lines of a given source of light by using the diffraction grating in the normal incidence position has been calculated and verified with standard values.

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